

Product Requirements Document (PRD)

Enterprise-Grade Algorithmic Trading System with Domingo with Agentic AI Assistant

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Introduction

This Product Requirements Document (PRD) outlines the specifications for an enterprise-grade algorithmic trading system designed to revolutionize how traders interact with financial markets. The system integrates a sophisticated Agentic AI Assistant as its central feature, enabling natural language interactions for trading, research, and portfolio management. Built on a "Best-of-Breed" integration strategy, the platform leverages leading open-source technologies to create a modular, scalable, and powerful foundation.

The architecture consists of microservices communicating via a centralized API bridge, with key components including a core trading engine, advanced data analytics, AI-driven strategy development tools, and a responsive user interface. This PRD details the functional and non-functional requirements, user stories, test cases, and development roadmap to guide the creation of this cutting-edge platform.

Why This Project Exists

This project is driven by a vision to transform algorithmic trading by making sophisticated tools accessible to a wide range of users, from novice traders to seasoned professionals. It is not merely another trading platform; it is a comprehensive system designed to empower users with precision, confidence, and innovation in navigating the complexities of financial markets.

The high-level "why" of this project includes:

- **Democratization of Trading Tools:** To provide access to advanced trading strategies and technologies traditionally reserved for institutional investors.
- **Innovation through AI:** To harness artificial intelligence and machine learning to deliver intuitive, powerful, and user-friendly trading capabilities.
- **Flexibility and Scalability:** To create a platform that adapts to the evolving financial landscape and scales with user needs.
- **User Empowerment:** To bridge the gap between cutting-edge technology and practical trading applications, enabling anyone to succeed with the right tools.

This system challenges the status quo by pushing the boundaries of algorithmic trading, setting a new standard for what a trading platform can achieve.

Objectives

- **Primary Objective:** Build a scalable, secure, and intuitive algorithmic trading platform with an integrated Agentic AI Assistant to enhance trading, strategy development, and portfolio management.
- **Secondary Objectives:**
 - Leverage best-in-class open-source technologies for each component.
 - Ensure a modular architecture for maintainability and future enhancements.
 - Deliver a seamless user experience across web and mobile platforms.
 - Meet enterprise-grade security and compliance standards.

Scope

The system encompasses the following major components:

- A core trading engine supporting multiple asset classes and broker integrations.
- Data management and analytics for real-time and historical market data.
- AI-powered tools for strategy creation, backtesting, and optimization.
- An Agentic AI Assistant for natural language interaction and task automation.
- A unified, responsive user interface for web and mobile access.
- Enterprise-grade security, compliance, and operational features.
- Advanced options trading capabilities.

Functional Requirements

Core Trading Infrastructure

- **Multi-Broker & Market Connectivity:**
 - Integrate with brokers such as Interactive Brokers, Alpaca, and TD Ameritrade.
 - Support for the FIX protocol for institutional connectivity.

- Real-time connection health monitoring and seamless switching between paper and live trading accounts.
- **Trading Instruments:**
 - Support for stocks, ETFs, futures, options, forex, and cryptocurrencies.
- **Order Management System (OMS):**
 - Basic order types: Market, Limit, Stop, Stop-Limit, Trailing Stop.
 - Advanced order types: GTC, IOC, FOK, VWAP, TWAP, Iceberg, OCO.
 - Order execution logging and robust error handling for failed orders.

Data Management & Analytics

- **Data Feed Management:**
 - Integrate free data sources (e.g., Yahoo Finance, Alpha Vantage).
 - Real-time trading data via Interactive Brokers.
 - Data resiliency with fallback mechanisms and real-time quality scoring.
- **Alternative & Advanced Data Integration:**
 - Real-time news feeds with sentiment analysis.
 - Fundamental, economic, and ESG data streams.
 - Exotic data sources (e.g., satellite imagery, IoT sensor networks).
- **Market Analytics & Microstructure:**
 - Advanced market screener with natural language filters.
 - AI-powered opportunity detection and cross-asset screening.
 - Market microstructure analysis for order book dynamics and liquidity.
- **Post-Trade Analytics:**
 - Transaction Cost Analysis (TCA), fill rate analysis, and broker performance evaluation.

AI-Powered Strategy Development & Backtesting

- **Strategy Creation Studios & Management:**

- No-code strategy builder with a drag-and-drop interface.
- Generative AI assistant to convert natural language prompts into strategy code.
- Python/API studio with cloud-based Jupyter notebooks for advanced users.
- **Backtesting & Optimization Engine:**
 - Enterprise-grade backtester with realistic modeling (slippage, commissions, market impact).
 - Advanced analytics: Monte Carlo stress testing, walk-forward optimization.
 - AI-assisted optimization with hyperparameter tuning and overfitting detection.
- **Strategy Management:**
 - Git integration for version control and collaboration.
 - Strategy marketplace for sharing and licensing.

Integrated Agentic AI Assistant

- **Core Intelligence: Agentic Framework:**
 - Agentic Retrieval-Augmented Fine-Tuning (RAFT) system.
 - Pre-tuned domains for finance and legal systems.
 - Custom fine-tuning with user-uploaded datasets.
 - Intelligent model routing and continuous learning pipeline.
- **Document & Content Processing:**
 - Universal ingestion of file formats (text, PDFs, spreadsheets, images, audio/video).
 - Intelligent processing pipeline with noise removal and content validation.
- **Database & Analytics Integration:**
 - Support for relational (PostgreSQL), vector (pgvector/Qdrant), and time-series databases.

- Automated database selection and query optimization.
- **Advanced Hybrid Search & Retrieval:**
 - Multi-modal search (semantic, keyword, graph-based).
 - Personalized and federated search with web integration and fact-checking.
- **Collaborative Agent Network:**
 - Specialized agents (Analyst, Researcher, Risk Manager, Compliance) for task delegation.

User Experience & Multi-Platform Accessibility

- **Unified User Interface (UI):**
 - Personalized dashboards with drag-and-drop widgets.
 - Institutional-grade charting with TradingView integration.
 - Guided workflows and tooltips for new users.
- **Conversational Chatbot Interface:**
 - Modern chat interface with real-time messaging and file uploads.
 - Long-term memory and advanced conversational intelligence.
- **Mobile, Desktop & Web Access:**
 - Responsive web app built with Next.js, React, and TypeScript.
 - Native mobile apps for iOS and Android using React Native.
 - Progressive Web App (PWA) with offline capabilities.
- **Accessibility & Internationalization:**
 - Compliance with WCAG 2.1 accessibility standards.
 - Multi-language support with real-time translation.

Enterprise-Grade Security, Compliance & Operations

- **Advanced Security & Compliance:**
 - Zero-Trust Architecture with continuous verification.

- End-to-end encryption and container security scanning.
- **Regulatory Compliance & Auditing:**
 - Compliance with GDPR, HIPAA, and financial regulations.
 - Automated compliance reporting and immutable audit trails.
- **Risk Management:**
 - Real-time risk hub for pre-trade checks and circuit breakers.
 - Advanced risk analytics (VaR, correlation analysis).
- **User Management & Entitlements:**
 - Role-Based Access LocalLLM Access Control (RBAC) with custom roles.
 - Single Sign-On (SSO) integration with major identity providers.
- **Operational Monitoring & DevOps:**
 - Observability stack with Prometheus, Grafana, and ELK.
 - Infrastructure as Code (IaC) using Terraform and Ansible.
 - Automated CI/CD pipeline and disaster recovery strategies.

Options Trading

- **Comprehensive Options Trading:**
 - Support for basic, complex, and exotic options strategies.
 - Tools for Greeks calculation, payoff diagrams, and pricing models.
- **Enhanced Features:**
 - Visual strategy builders and risk/reward simulators.
 - Pricing engines (Heston, Monte Carlo, binomial models).
 - Volatility suite with surface constructors and skew arbitrage detectors.
 - Risk management tools for Greek exposure and margin forecasting.

Non-Functional Requirements

- **Performance:**

- Handle high-frequency trading with latency under 100ms for critical operations.
 - API response times below 100ms.
- **Scalability:**
 - Horizontal scaling to support growing user base and data volume.
 - Modular architecture for independent component scaling.
- **Security:**
 - Compliance with OWASP Top 10 security standards.
 - Regular audits and penetration testing.
- **Reliability:**
 - 99.99% system uptime with robust failover mechanisms.
 - Automated backups and disaster recovery plans.
- **Usability:**
 - Intuitive UI with a minimal learning curve.
 - Comprehensive documentation and user guides.

User Stories

- **As a novice trader**, I want a no-code strategy builder so I can create trading strategies without programming knowledge.
- **As an experienced trader**, I want a Python studio to develop and test complex strategies with full control.
- **As a portfolio manager**, I want to ask the AI assistant for real-time portfolio insights using natural language.
- **As a compliance officer**, I want automated reporting and audit trails to ensure regulatory compliance.
- **As a mobile user**, I want full platform functionality on my smartphone for trading on the go.
- **As an options trader**, I want tools to calculate Greeks and visualize payoff diagrams for my strategies.

Test Cases

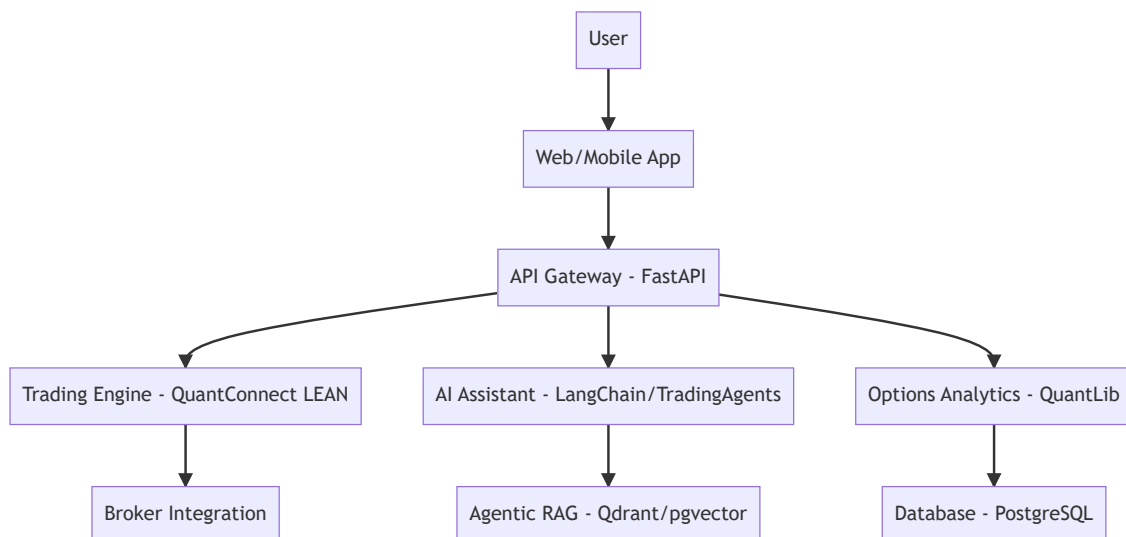
- **Test Case 1: Broker Connectivity**
 - **Description:** Verify the system connects to Interactive Brokers and executes a market order.
 - **Steps:** Connect to IB, place a market order for 10 shares of AAPL, confirm execution.
 - **Expected Result:** Order executed successfully, logged in the system.
- **Test Case 2: AI Assistant Query**
 - **Description:** Ensure the AI assistant processes a natural language query and returns portfolio data.
 - **Steps:** Ask, "What's my current portfolio value?" via the chatbot.
 - **Expected Result:** Accurate portfolio value returned in real-time.
- **Test Case 3: Backtesting Accuracy**
 - **Description:** Validate the backtesting engine with historical data.
 - **Steps:** Run a moving average crossover strategy on AAPL data from 2022.
 - **Expected Result:** Results match expected performance metrics (e.g., Sharpe Ratio).
- **Test Case 4: Risk Management**
 - **Description:** Check that pre-trade risk checks prevent invalid orders.
 - **Steps:** Attempt to place an order exceeding account risk limits.
 - **Expected Result:** Order rejected with an appropriate error message.
- **Test Case 5: UI Responsiveness**
 - **Description:** Confirm the UI is accessible and responsive on web and mobile.
 - **Steps:** Access dashboard on desktop and mobile, resize window.
 - **Expected Result:** UI adapts seamlessly without loss of functionality.

UI Mockups and Wireframes

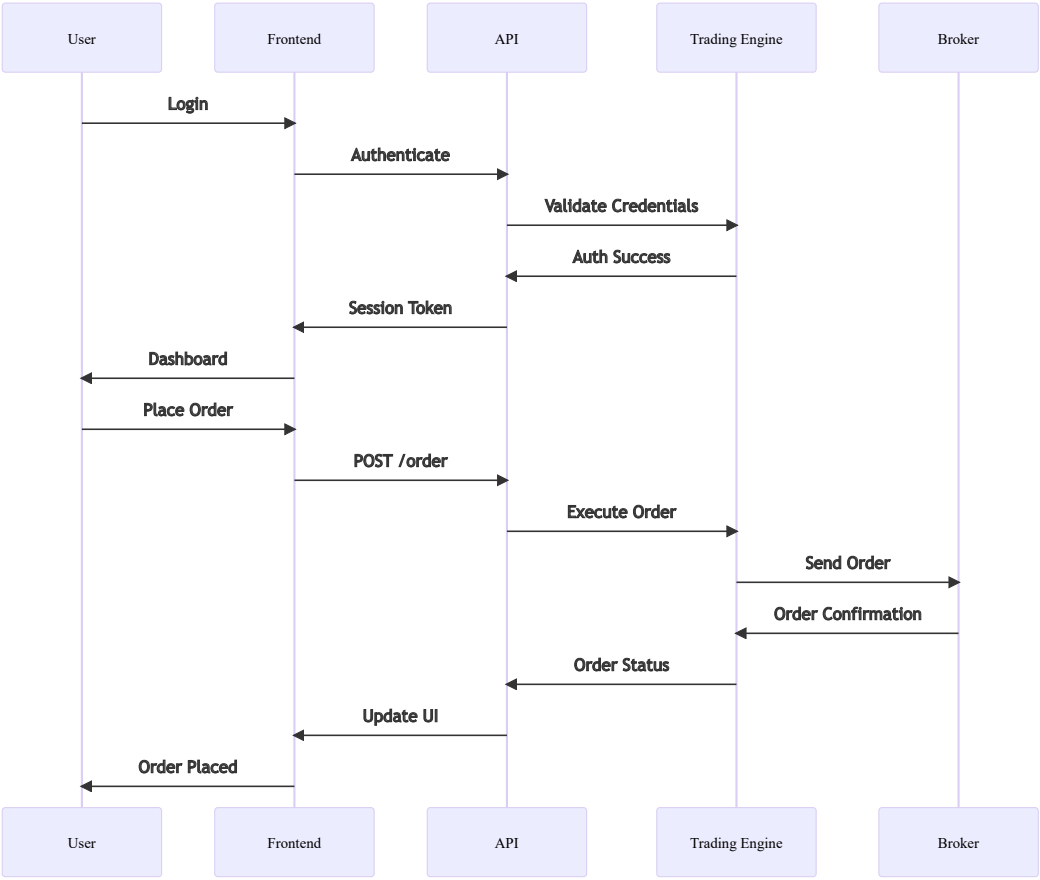
- **Dashboard Wireframe:**
 - A central dashboard with widgets for portfolio overview, recent trades, and market news.
 - Drag-and-drop functionality for widget customization.
- **Strategy Builder Mockup:**
 - Visual interface with blocks for indicators, conditions, and actions.
 - Real-time preview of strategy logic and performance metrics.
- **AI Chatbot Interface:**
 - Chat window supporting text input, file uploads, and voice commands.
 - Contextual suggestions and quick actions based on queries.

Diagrams

System Architecture



Order Execution Flow



Development Phases

The project follows a 5-phase roadmap:

1. Phase 1: Foundational Setup

- Objective: Establish core infrastructure with QuantConnect LEAN and PostgreSQL/pgvector.
- Key Deliverables: Dockerized environment, basic backtesting with Yahoo Finance data.

2. Phase 2: API Bridge

- Objective: Build a FastAPI wrapper for programmatic control of the trading engine.
- Key Deliverables: POST /backtest endpoint, integration of TA-Lib and pandas-ta.

3. Phase 3: AI Assistant MVP

- Objective: Integrate LangChain and TradingAgents for an initial AI assistant.
- Key Deliverables: Agentic RAG pipeline, tools for backtesting and document queries, Open-WebUI chat interface.

4. Phase 4: Full User Experience

- Objective: Develop the Next.js frontend and enable paper/live trading.
- Key Deliverables: TradingView charts, live trading endpoints, enhanced AI tools.

5. Phase 5: Advanced Features & Enterprise Readiness

- Objective: Add advanced analytics, security, and monitoring features.
- Key Deliverables: QuantLib integration, no-code builder, enterprise guardrails, observability stack.

Appendices

- Glossary:

- **Agentic RAG:** Retrieval-Augmented Generation system with AI agents for information processing.
- **Zero-Trust Architecture:** Security model requiring continuous verification of all requests.
- **References:**
 - [QuantConnect LEAN Documentation](#)
 - [LangChain Documentation](#)
 - [QuantLib Documentation](#)

This PRD provides a detailed blueprint for the development of an enterprise-grade algorithmic trading system, ensuring all requirements are clearly defined and actionable. The integration of open-source technologies with custom development will create a robust, innovative platform ready to meet the needs of modern traders.